

Model Immunization from a Condition Number Perspective

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Condition-number control enables selective model immunization

- Model Immunization from a Condition Number Perspective
- Amber Yijia Zheng, Cedar Site Bai, Brian Bullins, Raymond A. Yeh
- A condition-number view of making harmful fine-tuning difficult while preserving benign utility

Problem, Idea, and Results at a Glance

- Immunize models so harmful fine-tuning is difficult while benign utility is preserved
- Use the Hessian condition number as a proxy for fine-tuning difficulty
- Introduce differentiable regularizers with **monotone guarantees** and validate on linear and deep models

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Immunization is feasible when task covariances misalign

- Immunize against harmful adaptation without damaging benign performance
- Link convergence speed to the condition number to control fine-tuning effort
- Tie feasibility to alignment between harmful and pre-training covariances

Our method delivers monotone conditioning control

- Provide a condition-number framework for selective model immunization
- Develop differentiable regularizers that **reliably** increase or decrease conditioning
- Deliver a gradient-based algorithm for linear and deep settings with **clear update rules**

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Condition number governs fine-tuning speed; Rwell improves it

- Condition number governs gradient descent speed for strongly convex losses
- Rwell reduces conditioning **monotonically** under gradient steps
- Linear probing trains a head on frozen features to assess representation conditioning

Theory Intuition: Why kappa Controls Fine-tuning Speed

- Smaller conditioning leads to faster and more stable convergence
- Shaping the representation effectively preconditions the probing Hessian
- Decrease conditioning on benign data and increase conditioning on harmful data

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Immunized features hinder harmful probing while preserving benign utility

- Make harmful linear probing harder than using raw inputs
- Make benign linear probing no harder than using raw inputs
- Preserve benign-task performance when the mapping is invertible

Scope and Assumptions

- Evaluate immunization via linear probing with feature-induced Hessians
- Assume positive semidefinite covariances and unique extremal singular values for gradients
- Use step sizes that ensure **monotone** conditioning changes per regularizer

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Alignment determines whether we can shape harmful vs. benign conditioning

- The probing Hessian depends on the representation and the data covariance
- Alignment between benign and harmful covariances governs selective control
- Perfect alignment blocks simultaneous conditioning control **across tasks**

Objective: Jointly Shaping kappa on D_P and D_H

- Optimize a loss with Rwell on benign data and Rill on harmful data
- Encourage better benign conditioning while worsening harmful conditioning
- Balance utility and resistance through **joint regularization**

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Our objective jointly shapes benign and harmful κ

- Optimize $L(D_P, \omega, \theta) + R_{\text{well}}(H_P(\theta)) + R_{\text{ill}}(H_H(\theta))$
- Lower κ on D_P while raising κ on D_H to balance utility vs. resistance
- Tuning λ_P, λ_H trades off benign utility and immunization strength

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Closed-form gradients and safe steps guarantee monotone κ shifts

- Compute closed-form gradients for Rwell and Rill via the chain rule
- Update the head by standard loss gradients on benign data
- Update the representation by a normalized combination with safe step sizes

Monotone kappa Guarantees for Rill/Rwell

- Rill increases conditioning under suitable steps and smoothness
- Rwell decreases conditioning under similar step constraints
- Monotonicity holds despite indirect updates to the Hessian through features

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RIR quantifies immunization; we evaluate via linear probing

- Relative Immunization Ratio grows when harmful conditioning rises and benign conditioning stays small
- For deep models, estimate probing Hessians from feature batches
- Report RIR and benign accuracy and visualize harmful convergence slowdowns

Datasets at a Glance

- Linear settings include House Prices and MNIST binary pairs
- Deep settings use ImageNet for benign and Cars or Country211 for harmful
- Models include ResNet18 and ViT with partial block immunization

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RIR increases when harmful κ_H rises and benign κ_B falls

- Large increase in κ_H with small or decreased $\kappa_B \rightarrow \text{RIR} \gg 1$
- Intuition: harmful fine-tuning slows while benign adaptation stays efficient
- We report RIR alongside benign accuracy to ensure utility is preserved

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Our surrogates outperform Opt κ with greater stability

- Differentiable surrogates (Rwell/Rill) provide **monotone** conditioning control with stable, smooth gradients
- Compared to Opt κ , our method attains comparable-or-better RIR with **lower variance**
- Single-baseline setup isolates surrogate vs. direct κ optimization under matched implementation

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Our surrogates deliver stronger, stabler immunization than Opt κ

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- Across linear and deep, our method yields strong immunization with preserved benign utility
 - Opt κ is weaker and less stable than our surrogates
 - Benign utility is largely preserved while harmful adaptation slows

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Deep models inherit linear-probing guarantees empirically

- Estimate probing Hessians from feature matrices via grouped batches
- Immunize last blocks; evaluate via linear probing on D_H and D_P
- Observed κ trends and RIR match linear intuitions empirically

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Immunization complements unlearning and classic preconditioning

Positioning: Unlearning vs. Immunization and Conditioning

- Immunization targets prevention of re-finetuning rather than data removal
- Our novelty is deliberate increase of harmful-task conditioning
- This framing complements classic preconditioning that reduces conditioning

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Ablation: surrogate weights trade off utility vs. immunization

- Increasing λ_H raises κ_H faster but can hurt D_P if λ_P is too small
- Larger λ_P stabilizes benign utility and smooths representation updates
- Balanced weights yield monotone control with low-variance RIR

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Broader Related Work Touchpoints: Optimization Conditioning and Preconditioning

- Preconditioning reduces conditioning to accelerate and stabilize training
- Layer-wise and stochastic preconditioners improve deep optimization
- Our approach decreases benign conditioning and increases harmful conditioning for selective control

Limitations and Outlook

- Per-regularizer guarantees require careful tuning of weights and steps
- Theoretical guarantees cover linear features while deep results are empirical
- A single backward pass with a dummy layer simplifies implementation

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Limitations: guarantees are per-regularizer; deep results empirical

- Monotone κ shifts guaranteed per-regularizer for $H(\theta) = \theta^{\top K \theta}$
- Joint objective requires tuning; no global optimality guarantee
- Deep-model findings are empirical; benign accuracy may change

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Open Questions and Future Experiments

- Expand harmful benchmarks and assess cross-domain transferability
- Stress-test robustness against attacker adaptations and editing
- Move beyond linear probing and automate tuning with adaptive steps

Takeaways and Impact

- Condition-number control enables selective immunization of harmful tasks
- Rwell and Rill provide **reliable** monotone levers for conditioning
- Experiments show strong immunization with preserved or **improved** benign utility

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Setup details: linear probing choices impact stability

- House Prices and MNIST: multiple seeds and zoning/digit splits ensure robustness
- Learning rates/epochs per task; weights balanced by gradient norms
- Report averages over seeds; avoid tables—present qualitative protocol notes

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Setup details: deep-net immunization respects pretraining structure

- Start from ImageNet-pretrained models; immunize final blocks only
- Train with SGD+momentum; defaults from ImageNet applied
- Compute κ via feature-based Hessians using grouped batches

- RIR computed from small feature batches flags harmful-task re-adaptability

Thank You

Questions & Discussion